Categories used by *Shaping the Safety Boundaries* (Gao et al., 2024)

Three prompt categories (§3.2 Experimental Settings):

- **Benign** — curated subset of **Alpaca** (Taori et al., 2023). Used because it is pre-filtered to contain only safe instructional content.
- **Harmful** — direct harmful prompts pulled from **RedEval** (Bhardwaj & Poria, 2023) plus **AdvBench** (Zou et al., 2023). Plain harmful requests an aligned model is expected to refuse.
- **Jailbreak** — generated by running **seven** automated jailbreak attack methods on AdvBench prompts: GCG-Individual, GCG-Universal, AdvPrompter, COLD-Suffix, AutoDAN, PAIR, ArtPrompt.

Total: 32,507 samples. Activations are taken as the **last-token vector** of the input, projected via **MDS** (not PCA / t-SNE). Backbone: **Vicuna-7B-v1.3**, with extensions to Llama-2-7B-Chat, Qwen-1.5-0.5B-Chat, and Vicuna-13B-v1.5.

Differences vs. the dataset we are building (Module 1 + Module 2)

Axis	Gao et al. - Safety Boundaries	Ours - Module 1 + Module 2
Category names	benign / harmful / jailbreak	benign / harmful_direct / jailbreak_wrapped
Benign source	Alpaca (instruction-following data)	WikiText-103 (raw encyclopedic text)
Harmful source	RedEval + AdvBench only	AdvBench + HarmBench + JailbreakBench + SaladBench
Jailbreak source	7 automated optimizers (GCG, AutoDAN, PAIR, AdvPrompter, COLD-Suffix, ArtPrompt, GCG-Universal) on AdvBench	TrustAIRLab human-authored in-the-wild templates, filled with our harmful-request pool
Label	Category tag - prompt-type, not behavior verified	Behavior-grounded - GPT-4 judge with tau=7 rubric on Gemma's actual response; label 1 only if substantively harmful
Judge	None for labeling. GPT-4o-mini used only for Just-Eval utility scoring	GPT-4 (Azure or OpenAI) 3-step decision tree producing binary jailbreak label
Activation pooled	Last token of the input	Mean of last 5 prompt tokens
Layers studied	All layers 0-31, focus on 4 / 15 / 30	Fixed set {5, 10, 15, 20, 25} of Gemma's 26-block stack
Projection / analysis	MDS (preserves global geometry)	PCA -> K-means -> DBSCAN -> UMAP
Backbone model	Vicuna-7B-v1.3, Llama-2-7B-Chat, Qwen-1.5-0.5B, Vicuna-13B	Gemma-2-2b-it (single model)
Dataset size	32,507 samples total	~11k prompts (4k benign + 2k harmful_direct + 5k jailbreak_wrapped)
Goal of dataset	Visualize safety boundary + train ABD inference-time defense that penalizes outlier activations	Train CVAE generator + reward model + clustering detector to map the jailbreak subspace from benign-only corruption

Conceptual differences

1. What "jailbreak" means in the data

Gao et al. use *machine-optimized adversarial suffixes* — outputs of GCG/AutoDAN/PAIR — which produce specific, often non-semantic token strings. Our jailbreak_wrapped uses *human-crafted social-engineering wrappers* from TrustAIRLab (DAN-style personas, hypothetical framings). The activation geometry these two families induce is genuinely different: optimizer attacks push activations along narrow gradient directions, while wrapped attacks shift them through semantic role-play.

2. Label semantics

In Gao et al. a sample is 'jailbreak' because it *came from* a jailbreak attack pipeline — the model's response is never inspected for the label. In our pipeline a sample is label-1 only if the GPT-4 judge confirms the response is substantively harmful. Our positive class is strictly smaller and behavior-verified, which matches our paper's §3.6 verifier (tau=7) and is needed to train the reward model R_phi that predicts whether a perturbation delta_f actually induces jailbreak behavior.

3. Benign distribution

Alpaca instructions are short, instruction-formatted, and topically diverse — close to the distribution of the harmful and jailbreak prompts (also instructions). WikiText is *out-of-distribution* for instruction-following: long encyclopedic prose with no request structure. The choice is deliberate in our paper: §3.3 uses WikiText specifically so the corruption pipeline operates on benign passages whose activations lie in a *non-instructional* region, and any jailbreak behavior elicited by delta_f injection is attributable to the perturbation, not to a residual harmful intent in the benign prompt.

4. End goal of the dataset

Gao et al. need labeled clusters to (a) visualize the safety boundary on MDS plots and (b) tune an inference-time penalty (ABD) via Bayesian optimization. We need labeled (benign, harmful) activation pairs to (a) warm-start the CVAE on a binary concept label, (b) train the reward model that scores whether a generated delta_f jailbreaks Gemma, and (c) build a prompt-agnostic detector from cluster centroids.

5. Coverage of jailbreak diversity

Theirs spans 7 *attack algorithms* on a single seed prompt set (AdvBench). Ours spans ~1.4k unique TrustAIRLab templates × multiple harmful payloads, capped at 20 fills/template — trading attack-algorithm diversity for template diversity within a single attack family (in-the-wild human wrappers).

Bottom line

Gao et al.'s dataset is **algorithm-attack-centric** (optimizer outputs labeled by source) and **instruction-distribution benign**. Ours is **template-attack-centric** (human-authored wrappers labeled by behavior) and **non-instructional benign**. Both produce three categories, but the categories cover different slices of the jailbreak space and serve different downstream objectives — their categories drive a defense that lives at inference time, ours drive a generative/clustering map of the latent jailbreak subspace.